

Ibadan Urban Heat Risk Intelligence System

Summary Report | 2025

Urban Heat Island intensity mapping, spatial autocorrelation, GWR heat-driver modelling, and thermal vulnerability assessment for Ibadan Metropolis, Oyo State, Nigeria.

Methodology Conceptualisation

- Landsat dry-season LST is used as the thermal exposure baseline.
- NDVI, NDBI, and GHSL built-up density are used as heat-driver indicators.
- Moran's I, LISA, and Getis-Ord Gi* identify spatial clustering of heat.
- GWR uses a 500 m grid and forced adaptive bandwidth for stable local modelling.
- HVI combines exposure, sensitivity, and adaptive capacity indicators.
- OSM green/water distance layers are treated as static adaptive-capacity proxies.

Key Raster Statistics

Land Surface Temperature: min=27.819, mean=34.829, max=48.602, std=3.453

Urban Heat Island Intensity: min=-5.277, mean=1.733, max=15.506, std=3.453

Normalized Difference Vegetation Index: min=-0.239, mean=0.484, max=0.830, std=0.150

Normalized Difference Built-up Index: min=-0.515, mean=-0.084, max=0.292, std=0.124

Heat Vulnerability Index: min=0.054, mean=0.282, max=0.782, std=0.122

Heat Exposure Index: min=0.000, mean=0.337, max=1.000, std=0.166

Sensitivity Index: min=0.000, mean=0.048, max=0.881, std=0.104

Adaptive Capacity Index: min=0.090, mean=0.480, max=0.885, std=0.131

Result Interpretation

- Higher LST, UHI, HVI, and hot-spot membership indicate priority heat-risk zones.
- Negative NDVI coefficients imply vegetation cooling effects.
- Positive NDBI and built-up density coefficients imply urban surface warming effects.
- LGA rankings support planning prioritisation but should be validated locally.

Referenced Output Tables

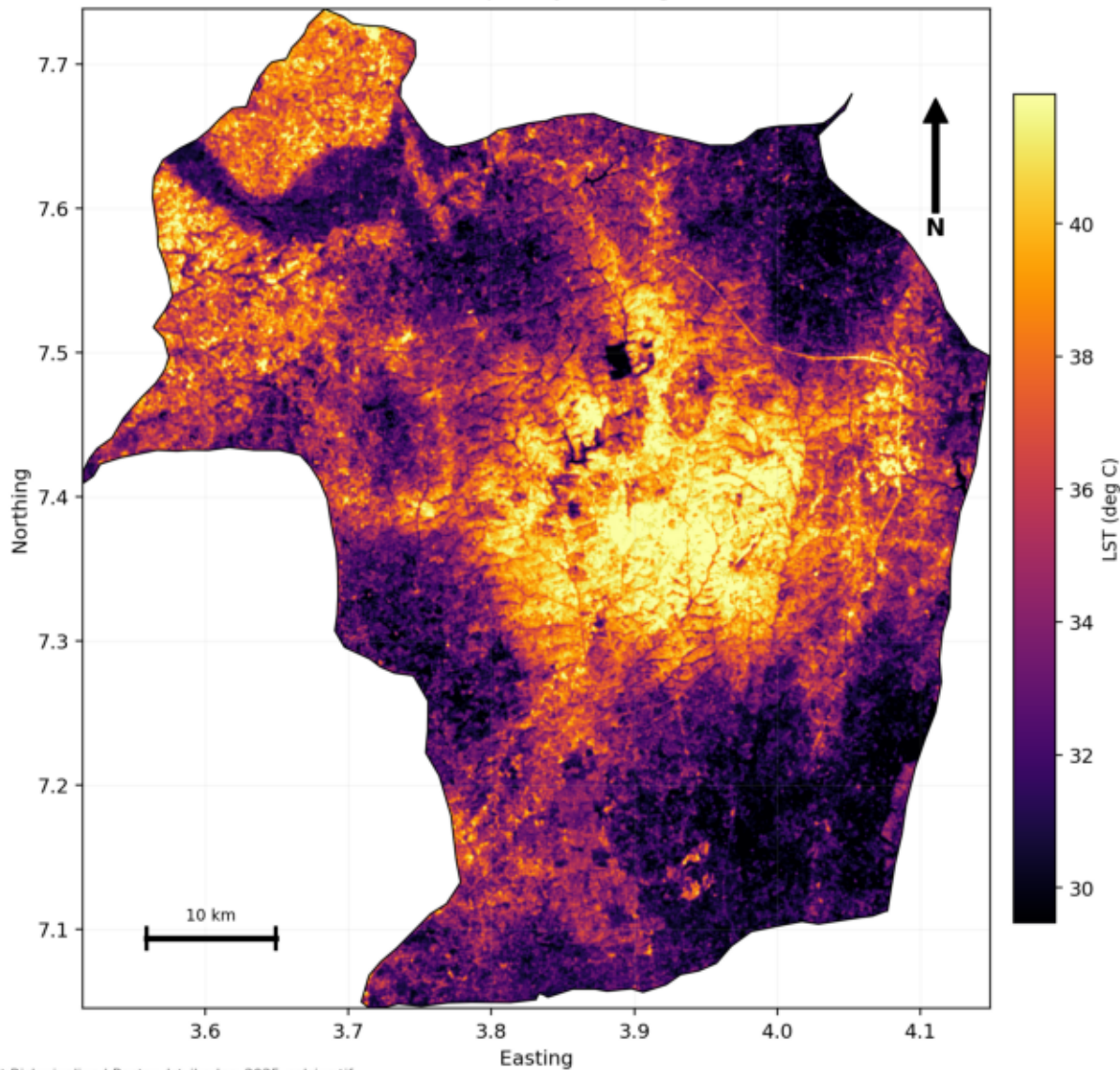
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Land Surface Temperature (2025)

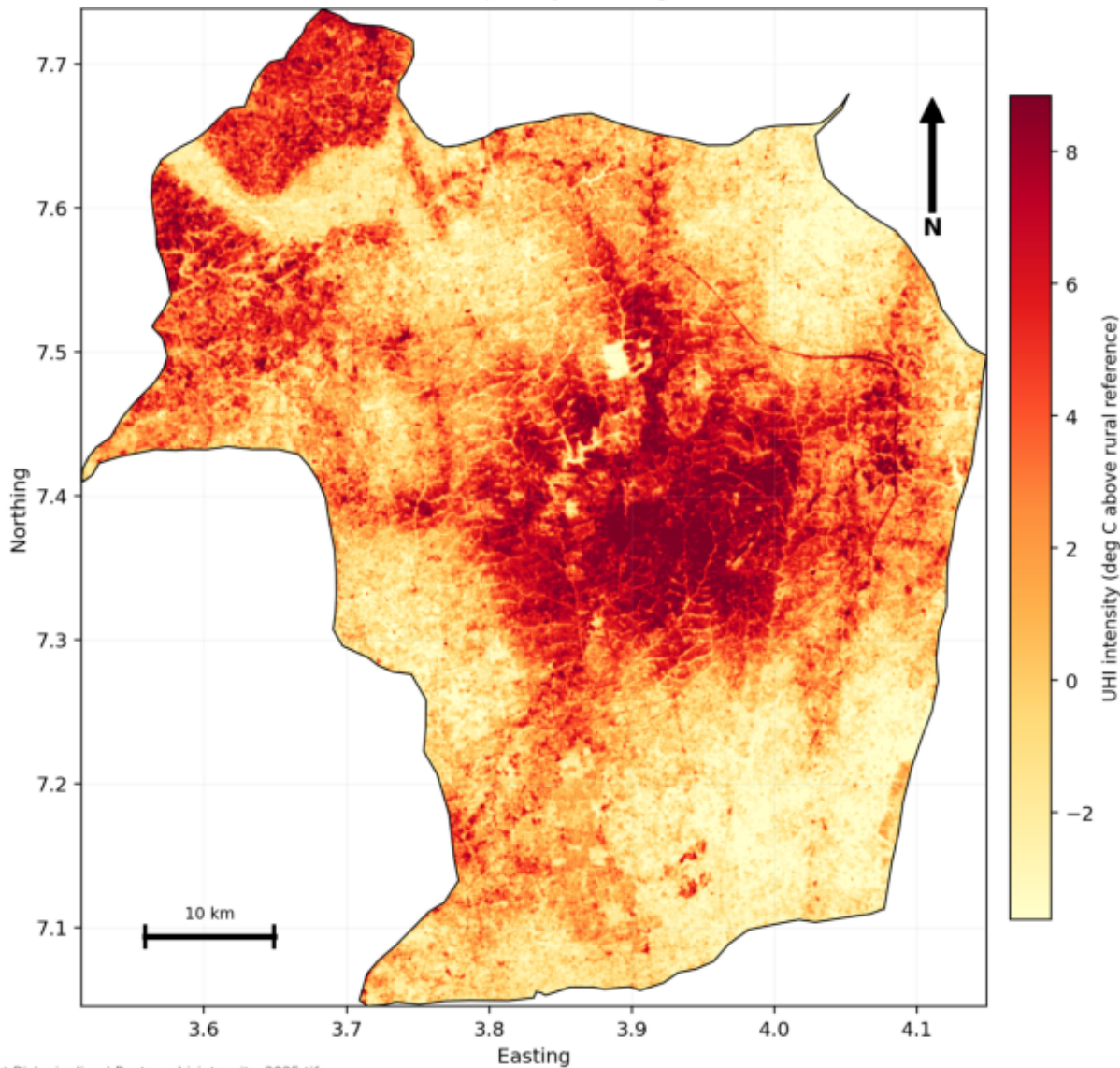
Ibadan Metropolis, Oyo State, Nigeria



Uhi

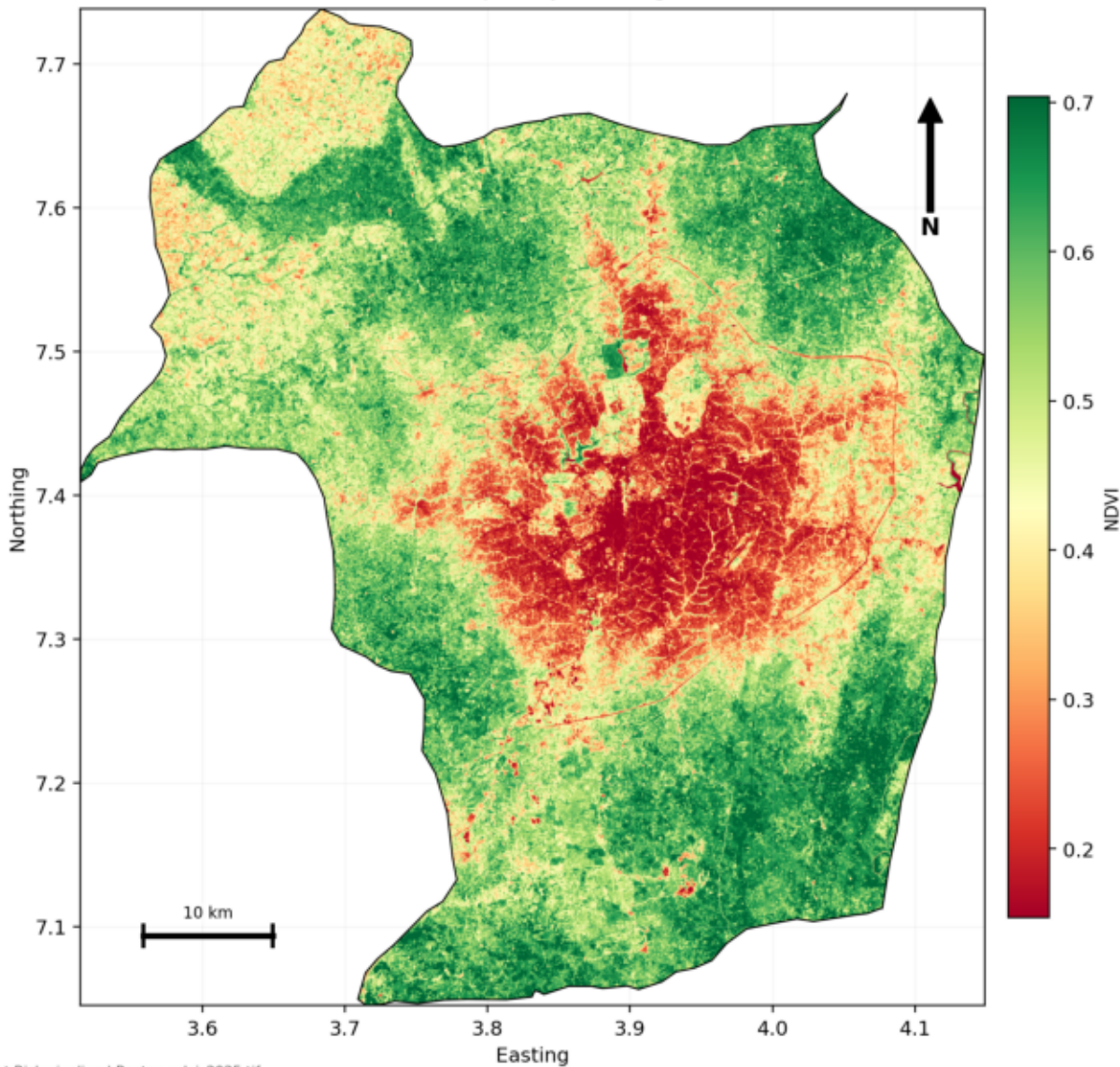
Urban Heat Island Intensity (2025)

Ibadan Metropolis, Oyo State, Nigeria



Ndvi

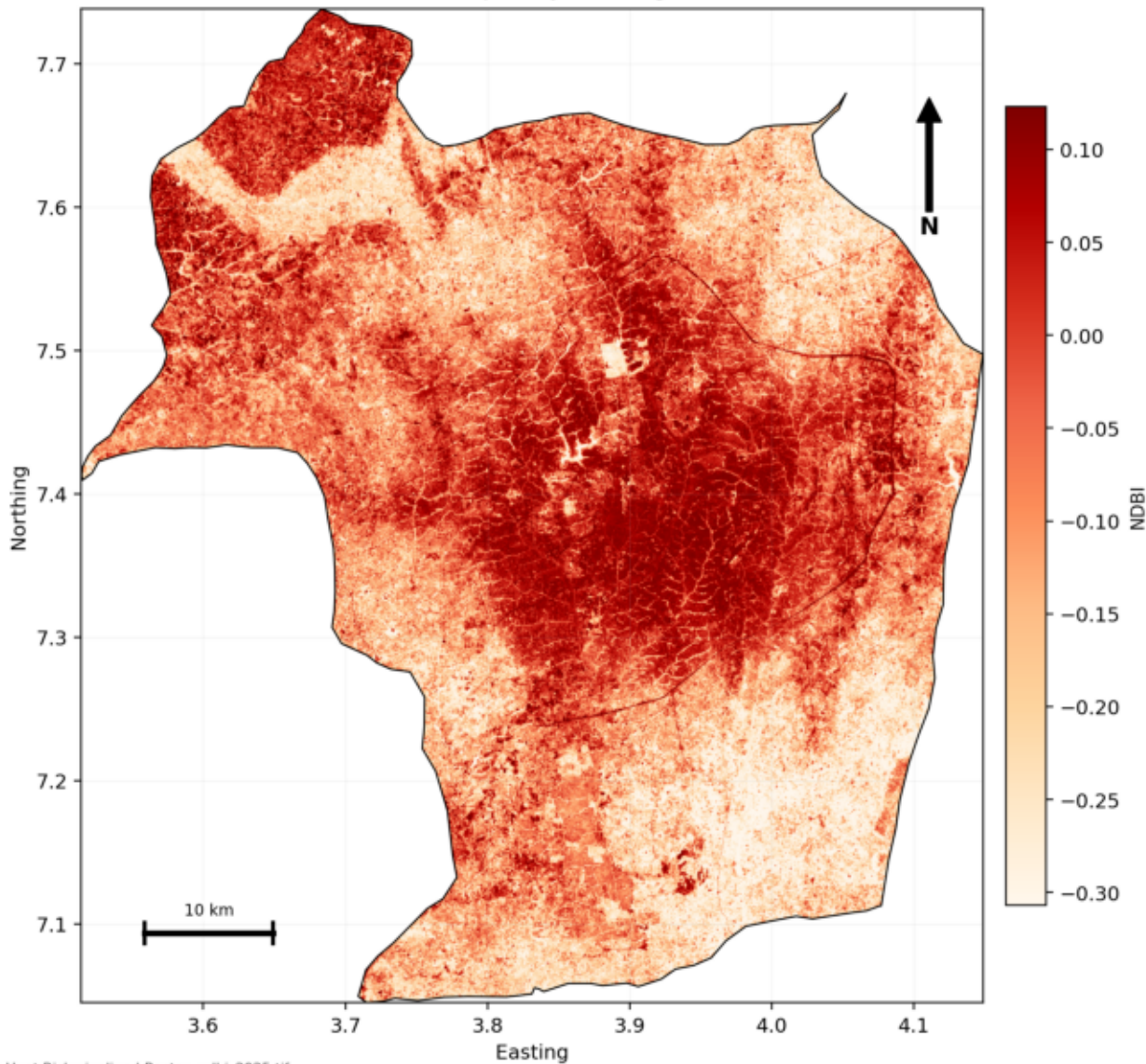
Normalized Difference Vegetation Index (2025)



Ndbi

Normalized Difference Built-up Index (2025)

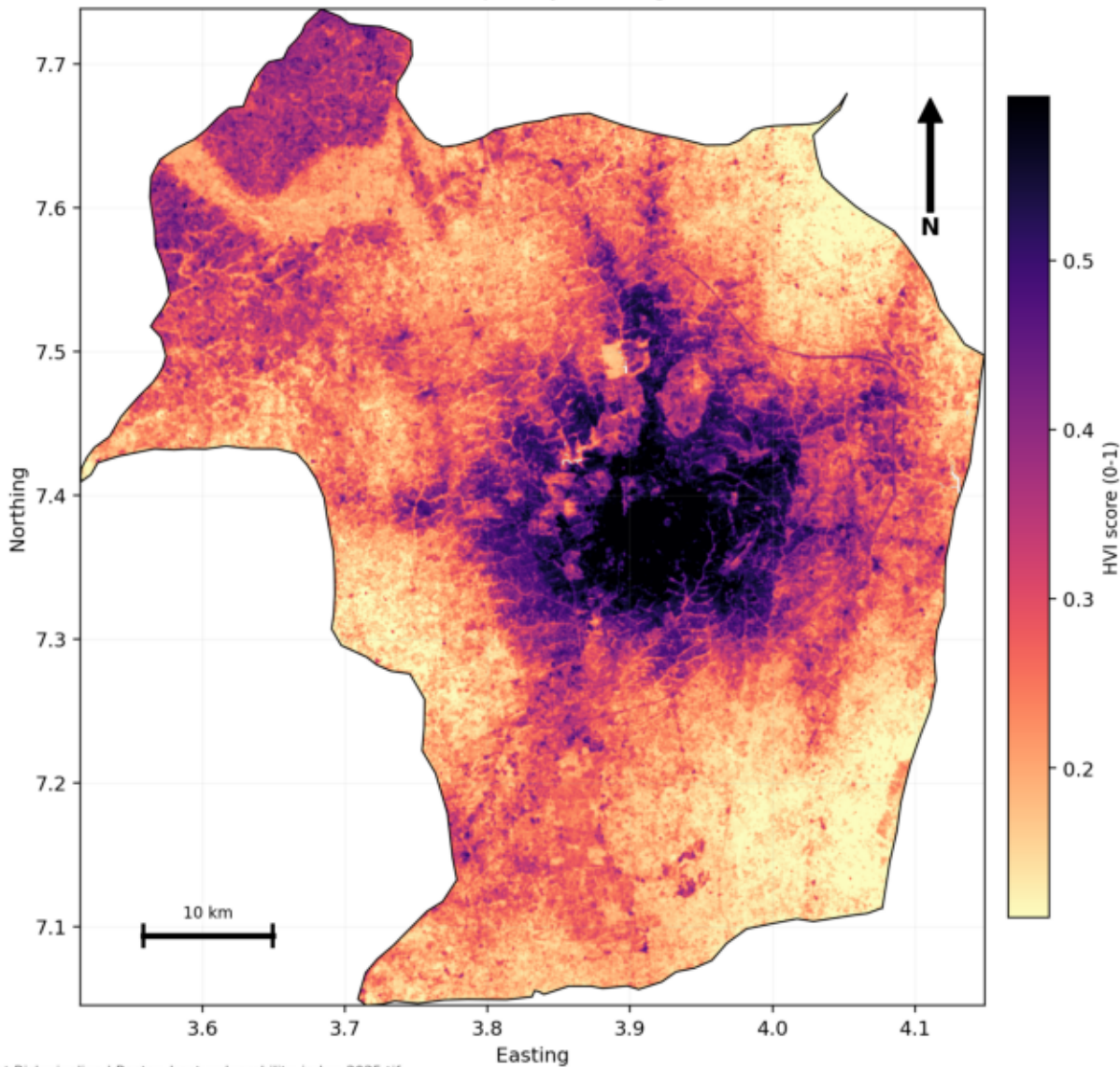
Ibadan Metropolis, Oyo State, Nigeria



Hvi

Heat Vulnerability Index (2025)

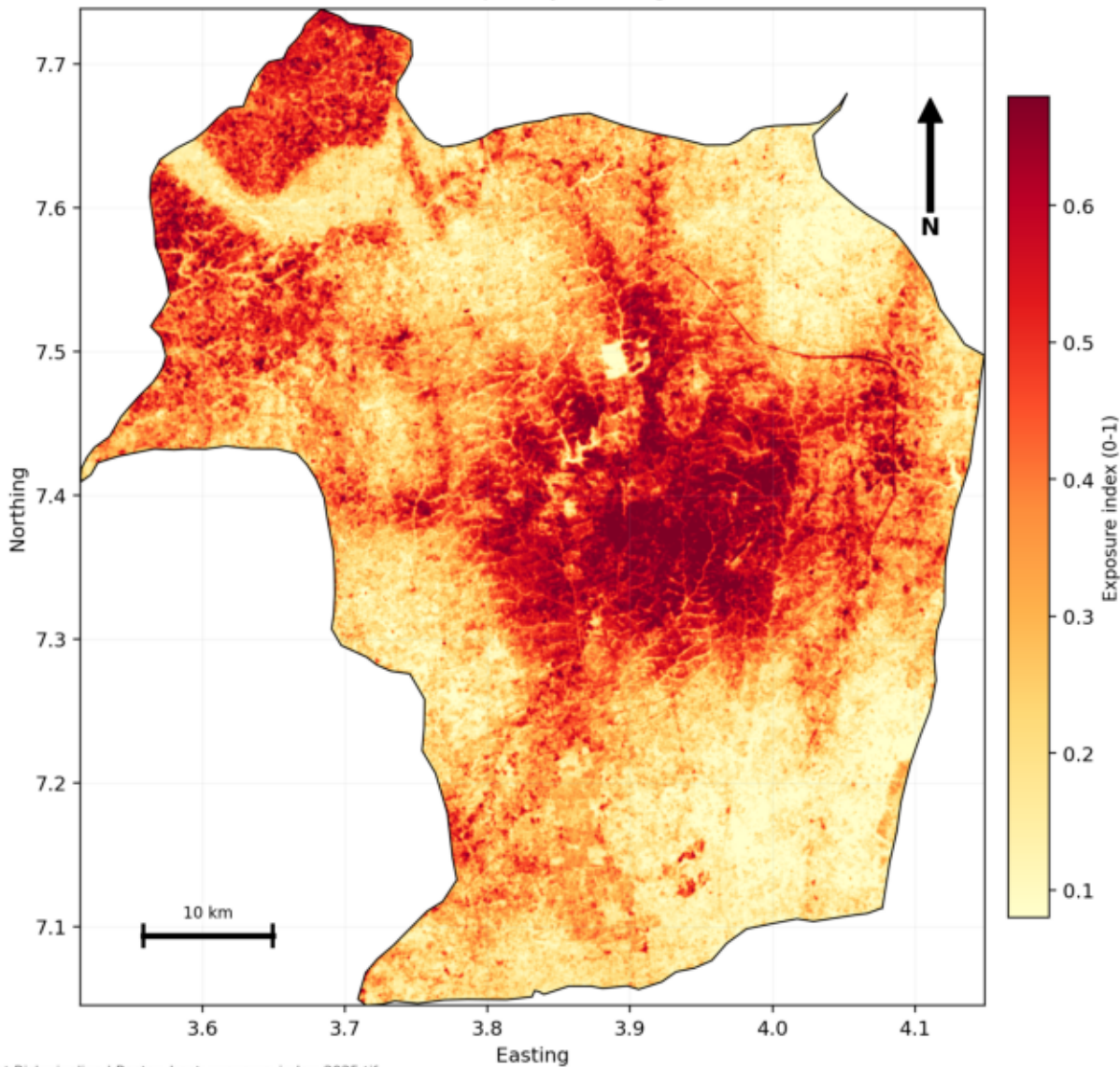
Ibadan Metropolis, Oyo State, Nigeria



Heat Exposure

Heat Exposure Index (2025)

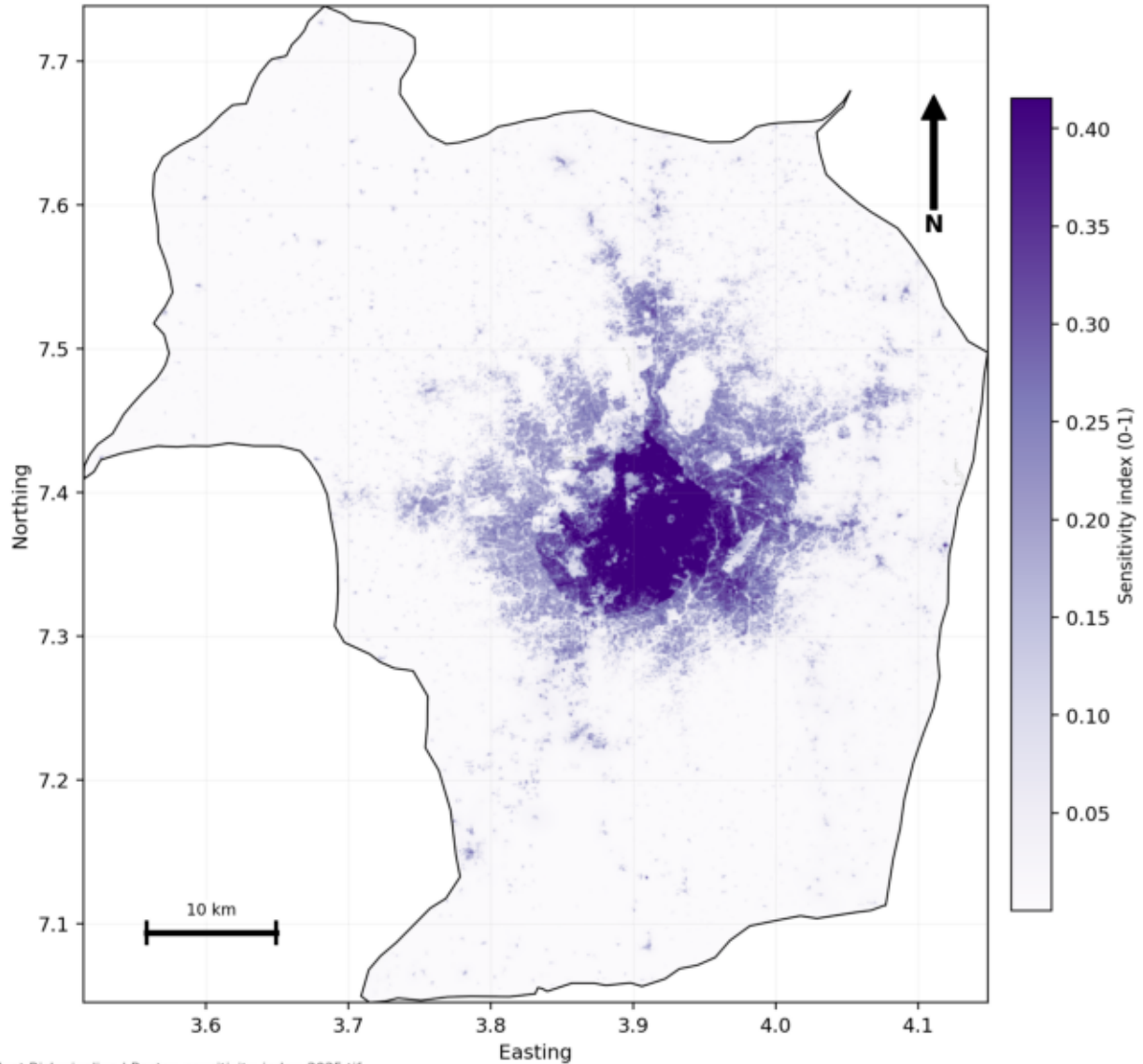
Ibadan Metropolis, Oyo State, Nigeria



Sensitivity

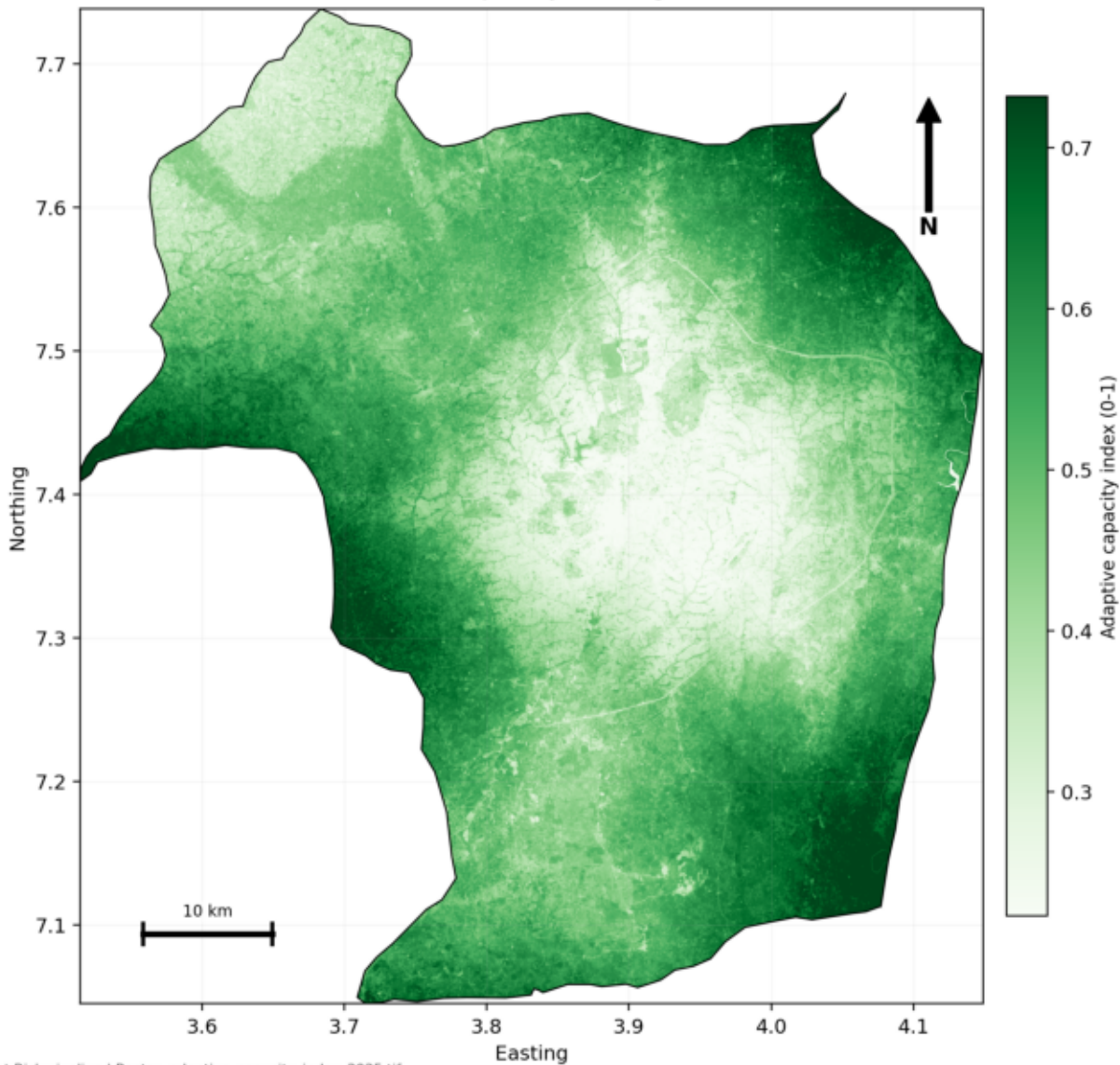
Sensitivity Index (2025)

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Adaptive Capacity

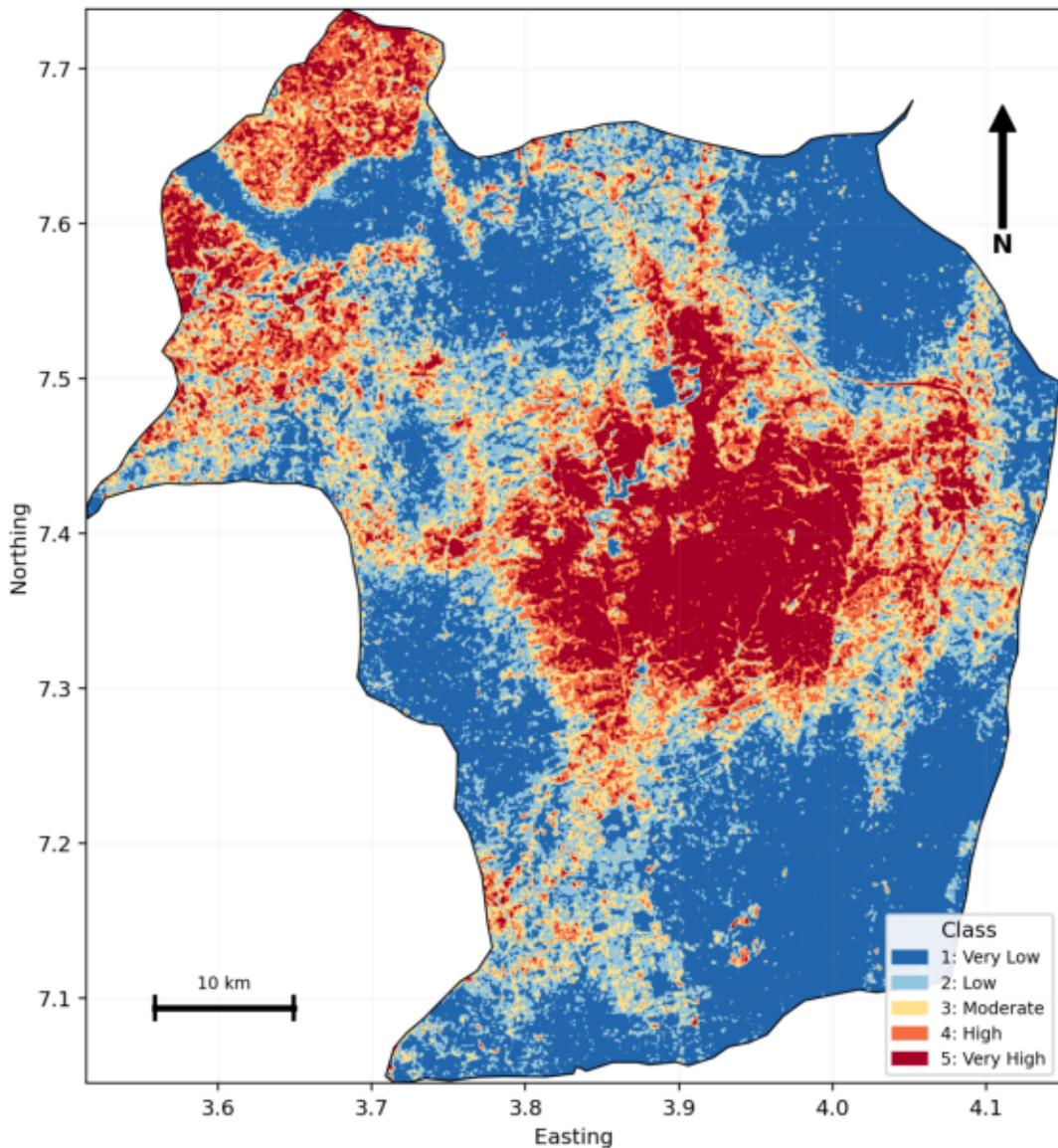
Adaptive Capacity Index (2025)



Uhi Classes

UHI Thermal Stress Classes (2025)

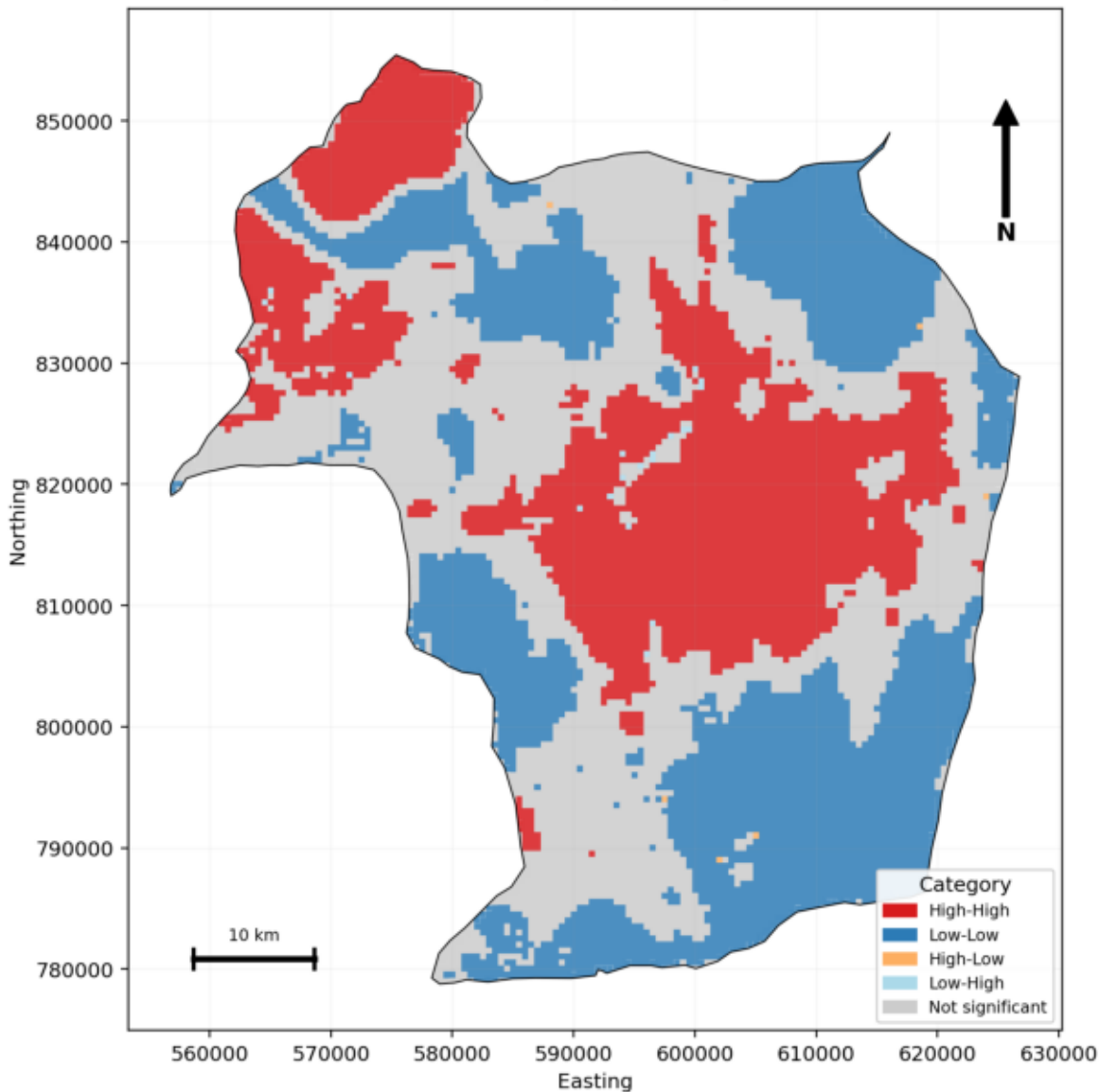
Ibadan Metropolis, Oyo State, Nigeria



Lisa Clusters

LISA Cluster Map (2025)

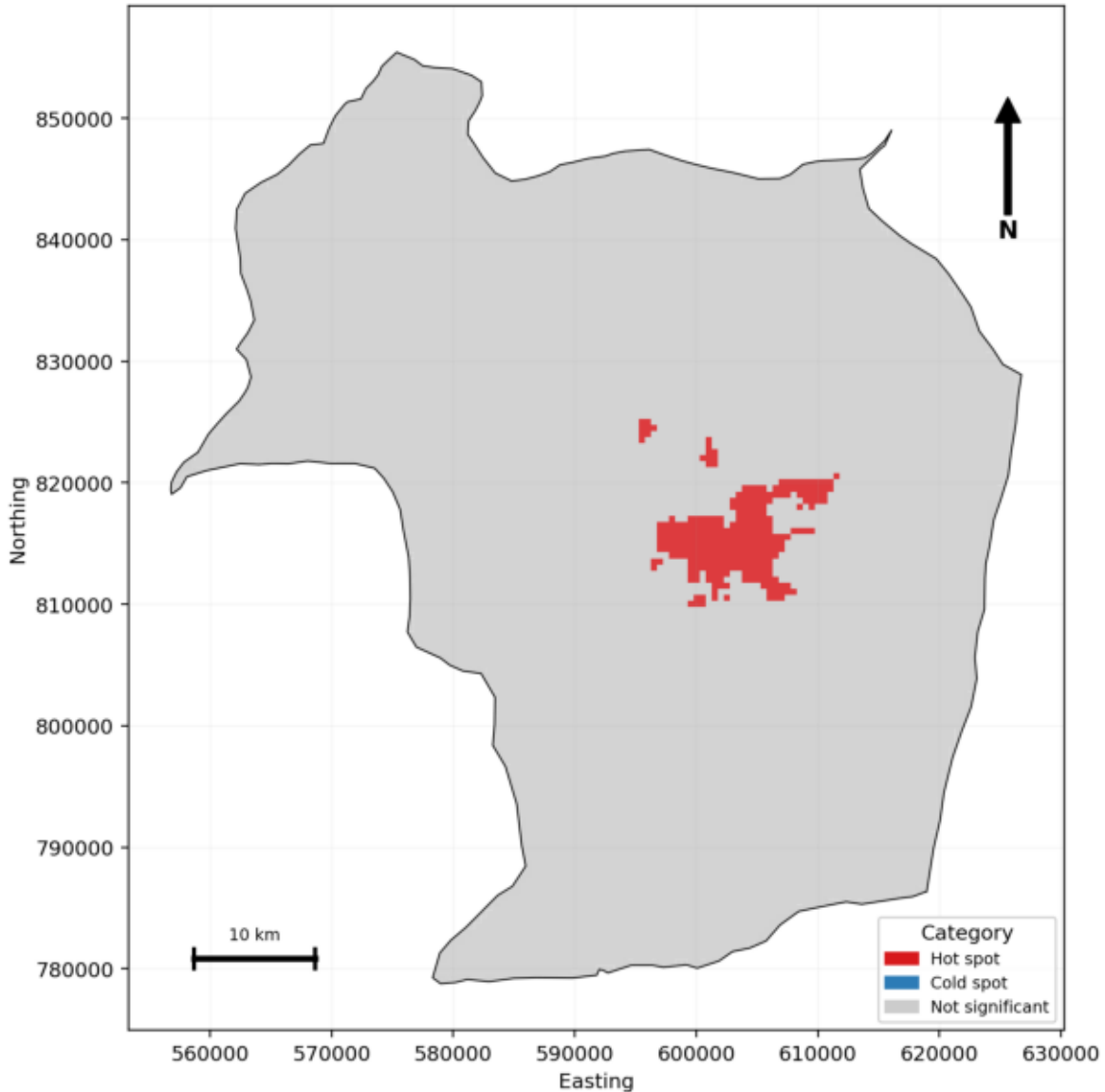
Ibadan Metropolis, Oyo State, Nigeria



Gistar Hotspots

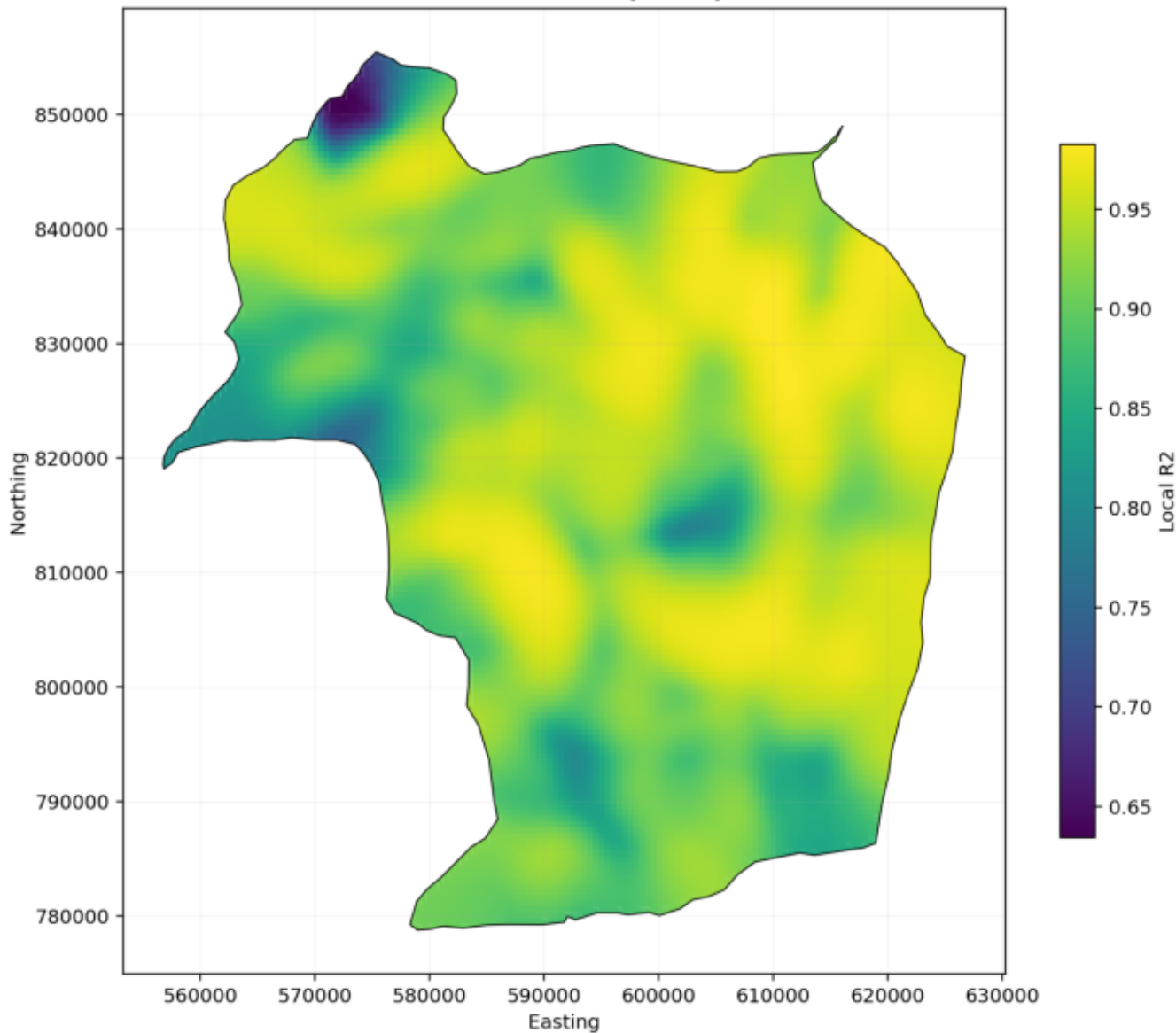
Getis-Ord Gi* Hot Spot Map (2025)

Ibadan Metropolis, Oyo State, Nigeria



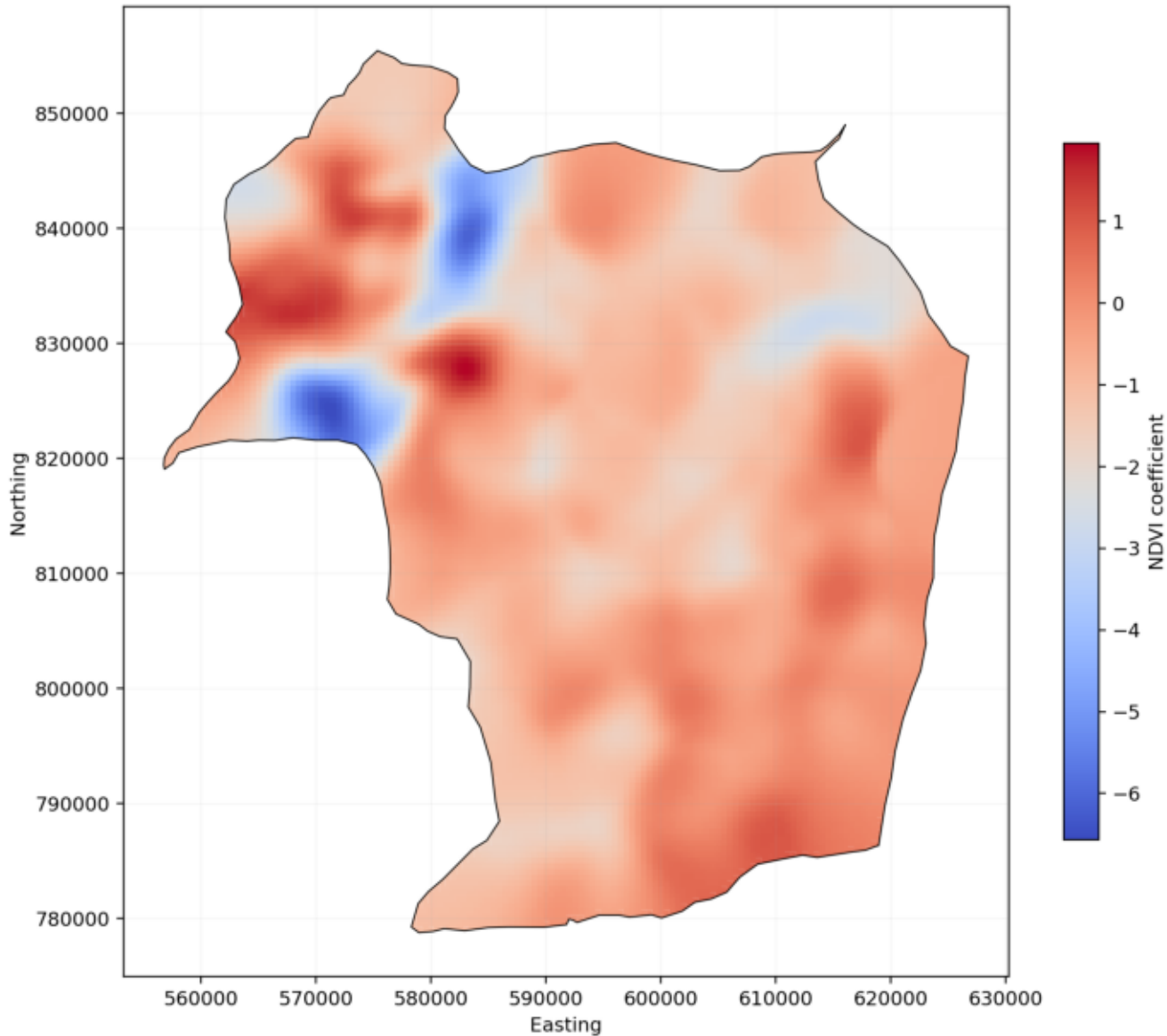
Gwr Local R2

GWR Local R2 (2025)



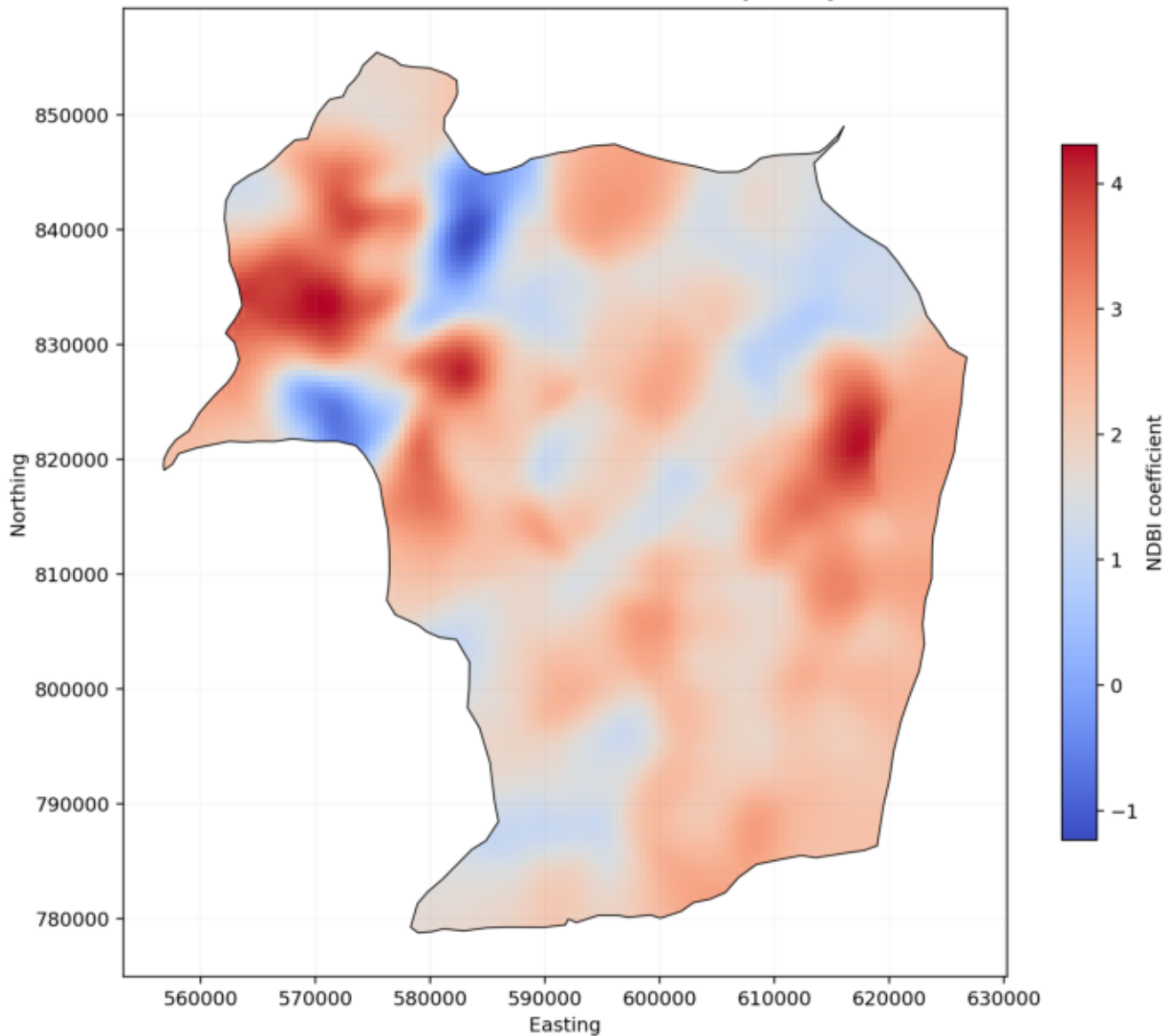
Gwr Coef Ndvi

GWR Local Coefficient: NDVI (2025)



Gwr Coef Ndbi

GWR Local Coefficient: NDBI (2025)



Gwr Coef Built Up Density

GWR Local Coefficient: Built-up Density (2025)

